

2025 – a Year of Interesting Properties

By Kevin Z.

2025 is a perfect square number!

In case you did not already know, 2025 is the 45th perfect square. Most people probably will not live through more than 1 perfect square year in their lifetime. For someone to have lived through both 44^2 and 45^2 , they would have to be at least 89 years of age – a feat accomplished by less than 30% of the human population. Even then, living to be older than 89 does not guarantee living through 2 perfect square years; in addition, one's birth year must be slightly smaller than a perfect square year for this rare event to take place (if someone was born in 2026, they would have to live to be 183 years old to see 2 perfect square years). For those who are still children in 2025, the odds of seeing

the next perfect square year – 2116 – are far from slim. Ever-improving medical technologies should make the human life-expectancy rise higher and higher as time goes on. In the future, people might even get the opportunity to live through 3 perfect square years!

2025 is the sum of the first 9 perfect cubes!

The sum of the first n perfect cubes can always be written as the square of a triangular number (a number which is the sum of all positive integers less than or equal to a positive integer n). Why is this the case? Using simple induction, it is easy to see that $1^3 + 2^3 + 3^3 + \dots + n^3 = (1 + 2 + 3 + \dots + n)^2$. Therefore, $1^3 + 2^3 + 3^3 + \dots + 9^3 = (1 + 2 + 3 + \dots + 9)^2 = 45^2 = 2025$.



2025 is the sum of the first 45 odd integers!

The sum of the first n odd integers can always be written as n^2 ; this can easily be proven through Gauss Sums. Thus, $1 + 3 + 5 + \dots + 89$, the sum of the first 45 odd integers, is equivalent to 2025.

New Year's Math Crossword

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See Answers on the Next Page!

ONE DOWN

At 12:00 AM on New Year's Day, Eric starts launching fireworks. In total, he launches 10 fireworks, where the k th and $(10 - k + 1)$ th fireworks have a radius of k meters ($1 \leq k \leq n$). Assuming that each firework is in the shape of a circle, find $A+1$ given that the average area of a firework can be written as $A\pi$.

ONE ACROSS

According to various polls, over 30% of all Americans set resolutions for the new year; however, much fewer than 30% actually end up sticking true to their resolutions. Let the probability of not failing one's new year's resolution on the x th day of January be modeled by the function $f(x)$ shown below.

$$\frac{1}{4} \left(\sin \left(\frac{2\pi}{31} (x + 7.75) \right) + 3 \right)$$

Given that there are n days where the probability of not failing one's new year's solution exceeds 75%, find n .

THREE ACROSS

If the function $P(x) = x^3 + (2a + 5)x^2 + 25ax + 2025$ has all integer roots, find a .

TWO DOWN

At a New Year's party, all the attendees shake hands twice with everyone else there. What is the least number of people that need to be at the party for the number of total handshakes to exceed 3100?

1.	2.
3.	



JANUARY 2025						
SUN	MON	TUE	WED	THU	FRI	SAT
29	30	31	1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	1